

Praneel Magapu

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EDUCATION

North Carolina State University

Aug 2023 – May 2027

Bachelor of Science in Computer Science, GPA: 3.5/4.0

Raleigh, NC

Relevant Coursework: Data Structures, Operating Systems, Software Engineering, Artificial Intelligence

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, TypeScript, Bash

Frameworks / Libraries: Flask, FastAPI, React, Tailwind CSS, NLTK, Pydantic

Tools / Platforms: Git, Docker, AWS (Elastic Beanstalk, CLI), Firebase, Supabase, Google Apps Script, Jenkins, GitHub Actions

AI & Data: LLM integration, RAG systems, Multimodal AI (TwelveLabs), NLP (Regex, NLTK), MongoDB, SQL

EXPERIENCE

Align Technology

May 2026 – Present

Software Engineering Intern

Raleigh, NC

- Redesigned the support-structure generation system within a large C++ pipeline for directly 3D-printed dental aligners, replacing a manual finishing step and improving optical quality on the final part while reducing material and production time.
- Engineered support geometry with pixel-level image processing to optimize clean separation between the printed support and the final appliance.

DevDynamics.ai

May – Aug 2025

Software Engineering Intern

Remote

- Engineered an LLM-powered intent extraction pipeline and automated processing workflows, reducing manual triage overhead across engineering teams.
- Built a hybrid regex and LLM classifier with Google Workspace APIs to automatically classify and track email thread timelines across 100+ active threads.
- Deployed a production AI-powered Gmail Auto-Labeler using Flask, AWS, and Supabase, adopted internally for day-to-day email management.

Liquid Rocketry Lab, NC State

Sep 2024 – May 2025

Data Engineer

Raleigh, NC

- Built a real-time telemetry pipeline for live rocket engine tests using Python and MongoDB across containerized microservices, capturing abort and breach events with sub-second latency.
- Implemented an observable fault-detection pattern for automated abort detection and breach logging, improving safety response reliability during high-frequency sensor ingestion.

North Carolina State University

May 2024 – Dec 2025

Research Assistant

Raleigh, NC

- Game2Learn Lab (Dr. Tiffany Barnes): built a production RAG system powering MerryQuery, a generative AI educational assistant, driving full-stack integration and iterating on UI/UX from live pilot feedback.
- Crowd Label Quality Control (Dr. Ed Gehringer): extended algorithms to detect unreliable crowd labels at scale using interval analysis and iterative aggregation, improving the integrity of large ML training datasets.

PROJECTS

Mendacia | Python, Flask, React, TypeScript, TwelveLabs API, LLM

Hack_NCState 2026

APIs, Tailwind CSS

- Built a full-stack multimodal media forensics platform generating structured forensic reports from raw video, detecting propaganda, narrative distortion, and cross-modal inconsistencies using TwelveLabs scene intelligence and LLM reasoning.
- Designed a cross-modal consistency engine comparing narrative claims against visual scene data with timestamped reasoning, plus a defensive adapter layer enforcing a stable UI contract across volatile backend responses.

Hiring Agent | FastAPI, Pydantic, React, Docker, pytest, GitHub Actions

Jul – Aug 2025

- Built an end-to-end resume-to-job matching engine with ontology-based skill normalization and hybrid keyword and LLM scoring with rationale generation, exposed via a REST API with structured Pydantic contracts.
- Containerized with Docker and validated via a pytest and GitHub Actions CI pipeline, designed for extensibility across job board integrations.

Mesh Alignment Engine | Python, Open3D, NumPy, Trimesh,

Mar – May 2026

Matplotlib

- Implemented an ICP-based surface mesh registration pipeline aligning 3D scans with sub-millimeter RMSE, applying a multi-resolution coarse-to-fine strategy to accelerate convergence on high-density meshes.
- Built a deviation heatmap visualization mapping per-point geometric divergence between aligned meshes, enabling spatial quality analysis analogous to treatment tracking in orthodontic scan pipelines.